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3. Asymptotics

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Asymptotics for large time

Consider again three connected medicine storage tanks where tank 1 has a leak. For large time, the tanks will decay to zero. This is because when we write our problem in matrix form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ the matrix $\mathbf{A}$ has negative eigenvalues.

In the previous problem, we saw that initially solutions may still grow. This is known as the transient regime. The transient regime often requires knowing the exact solution of the problem, and the behavior may depend on the initial conditions.

However, the long time solution of any problem of the form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ will generically be dominated by the eigenvalue with the largest real part. That is, if $\lambda_1 > \lambda_2 > \lambda_3$, with corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, the solution will look like

$$\mathbf{x} \sim ce^{\lambda_1 t} \mathbf{v}_1.$$

where c is a number that depends on the initial value and the eigenvector matrix.

Unfortunately, the output of MATLAB's "eig" function does not output the eigenvalues in any particular order. We will then use an algorithm to find the eigenvalue with largest real part. We will then compare the asymptotic solution to the exact solution to see how well it captures the decay of the solution.

Script ?

Save Reset MATLAB Documentation (https://www.mathworks.com/help/)

```
1 A = [ -2 1 0; 1 -2 1; 0 1 -1];
2 x0 = [0.5;0;0.5];
3 t = linspace(0,4,100);
4 % Store the eigenvectors and eigenvalues in matrices V and D respectively.
5 [V,D] = eig(A);
6
7 %Define I to be the index of the largest eigenvalue of A. That is the largest real part.
8 % Hint try using sort() or max() and look into the MATLAB documentation for find
9
10
11 [m, I] = max(diag(D));
12
13 % Defined lambda to be the maximum eigenvalue, and v to be the maximum eigenvector
14
15
16
17
18 lambda = D(I,I);
19 v = V(:,I);
20 %
21 C = inv(V)*x0;
22 c = C(I,1);
23 asymp = c*exp(lambda*t)*v(1,1);
24 % use the matrix exponential to find the solution for the given initial conditions
25 for m=1:length(t)
26     x(:,m) = expm(A*t(m))*x0;
27 end
28 %
29
30 plot(t,x(1,:))
31 hold on
32 plot(t,asymp)
33 legend('x','asymp')
34 set(gca,'fontsize',18)
35 xlabel('Time')
36 ylabel('Volume')
```

Run Script ?

Assessment: All Tests Passed

Submit ?

✔ V is correct

✔ Test 1 passed

✔ D is correct

✔ I is correct

✔ lambda is correct

✔ v is correct

✔ Exponential matrix used

✔ x is correct

3. Asymptotics

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